CNN-Based Rice Leaf Classification Project

Final Project Report - PRCP-1001-RiceLeaf

1. Introduction

This project focuses on developing a Convolutional Neural Network (CNN) model for rice leaf classification using transfer learning. The objective is to classify rice leaf images into different categories, which can help in early disease detection and crop management in agriculture.

2. Project Overview

The project implements a deep learning solution using MobileNetV2 architecture with transfer learning to classify rice leaf images. The model was trained on a dataset consisting of 119 images (96 training images and 23 validation images) across 3 different classes.

3. Methodology

3.1 Dataset Preparation

The dataset was organized into three classes of rice leaf images. Due to the limited size of the dataset (96 training images and 23 validation images), data augmentation techniques were crucial for improving model generalization and preventing overfitting.

3.2 Data Augmentation

Image augmentation was implemented using TensorFlow's ImageDataGenerator with the following transformations:

- Rescaling: Normalizing pixel values to the range [0, 1] by dividing by 255
- Shear Transformation: Applying 0.2 shear range for geometric variation
- Zoom: Random zoom up to 20% to simulate different viewing distances
- Horizontal Flip: Randomly flipping images horizontally

- Rotation: Random rotation up to 20 degrees
- Width and Height Shift: Randomly shifting images by 20% in both dimensions
- Validation Split: 20% of data reserved for validation

Why: Data augmentation artificially increases the effective dataset size, which helps reduce overfitting and improves the model's ability to generalize to new, unseen images. This is particularly important with small datasets like the one used in this project.

3.3 Model Architecture

The model architecture is based on MobileNetV2, a lightweight and efficient deep learning architecture pre-trained on ImageNet dataset.

3.3.1 Choice of MobileNetV2

MobileNetV2 was selected for this project for several key reasons:

- Lightweight Architecture: Efficient memory usage and fast inference, making it suitable for resource-constrained environments
- High Performance: Achieves excellent accuracy even on small datasets
- Transfer Learning: Utilizes pre-trained ImageNet weights (trained on 14 million images), eliminating the need to train from scratch
- Feature Extraction: The pre-trained layers capture general visual features that are transferable to rice leaf classification

3.3.2 Transfer Learning Implementation

Transfer learning was implemented using the following strategy:

- Base Model: MobileNetV2 pre-trained on ImageNet with input shape (224, 224, 3)
- Frozen Layers: The base model layers were frozen (base_model.trainable = False) to preserve learned ImageNet features
- Feature Extraction: The top layer (classification head) was removed (include_top=False)
- Custom Classification Head: Added Global Average Pooling layer, Dropout layer (0.5), and Dense layer with softmax activation for 3-class classification

Why: Freezing the base model prevents overfitting on the small dataset while allowing the custom classification head to learn rice leaf-specific patterns. This approach significantly reduces training time and improves performance compared to training from scratch.

3.4 Model Configuration

The final model configuration included:

- Image Size: 224 x 224 pixels
- Batch Size: 16 images per batch
- Optimizer: Adam optimizer
- Loss Function: Categorical cross-entropy
- Dropout Rate: 0.5 (50% dropout to prevent overfitting)
- Epochs: 20

4. Training Process

The model was trained for 20 epochs on the augmented training dataset. During training, the model processed batches of 16 images, adjusting weights through backpropagation. Validation was performed after each epoch to monitor the model's performance on unseen data.

5. Results

The MobileNetV2 transfer learning model achieved the following performance:

- Training Accuracy: ~77%
- Validation Accuracy: ~78% (0.7826)

The validation accuracy closely follows the training accuracy, indicating that the model generalizes well without significant overfitting. This suggests that the data augmentation strategy and dropout regularization were effective in preventing overfitting on the small dataset.

6. Challenges Faced and Solutions

6.1 Small Dataset Size

Challenge: The dataset consisted of only 119 images (96 training, 23 validation), which is insufficient for training a deep learning model from scratch.

Solution: Implemented comprehensive data augmentation techniques to artificially increase the dataset size and applied transfer learning using MobileNetV2 pre-trained on ImageNet. This approach allowed the model to leverage features learned from millions of images while adapting to the rice leaf classification task.

6.2 Model Tuning and Overfitting

Challenge: During the development process, attempts were made to improve model accuracy by adjusting hyperparameters and increasing the number of training epochs.

Experiment: The model was retrained with modified hyperparameters and increased epochs to 40, expecting an improvement in accuracy.

Result: Contrary to expectations, increasing the epochs to 40 and modifying other hyperparameters resulted in a significant drop in validation accuracy from 78% to 38%. This indicates that the model began overfitting to the training data, memorizing patterns specific to the training set rather than learning generalizable features.

Solution: The original configuration with 20 epochs was retained, which achieved the optimal balance between learning and generalization. The final model with 78% validation accuracy represents the best performance achieved while maintaining good generalization capabilities.

Lesson Learned: More training epochs do not always lead to better performance, especially with small datasets. It is crucial to monitor validation metrics and implement early stopping to prevent overfitting. The optimal model configuration balances learning capacity with regularization techniques.

7. Why I Did This Project

7.1 Agricultural Impact

Rice is a staple food for billions of people worldwide. Early detection of rice leaf diseases is crucial for maintaining crop health and ensuring food security. Traditional manual inspection is time-consuming, labor-intensive, and requires expert knowledge.

7.2 Technology Application

This project demonstrates how deep learning and computer vision can automate the classification of rice leaf conditions. By using transfer learning, we can develop effective models even with limited datasets, making this technology accessible for agricultural applications in resource-constrained environments.

7.3 Learning Objectives

The project provided hands-on experience with:

- Transfer learning and pre-trained models
- Data augmentation techniques
- Hyperparameter tuning and model optimization
- Overfitting prevention strategies
- Working with small datasets

8. Conclusion

This project successfully developed a CNN-based rice leaf classification model using MobileNetV2 transfer learning, achieving 78% validation accuracy. The implementation demonstrates the effectiveness of transfer learning and data augmentation for handling small datasets in agricultural applications.

The project highlighted the importance of careful hyperparameter tuning and the risks of overfitting, particularly when working with limited data. The final model configuration strikes an optimal balance between model capacity and generalization, providing a foundation for future improvements and real-world deployment.

9. Technical Details

Technology Stack:

- TensorFlow and Keras for deep learning model development
- MobileNetV2 architecture (pre-trained on ImageNet)
- ImageDataGenerator for data augmentation
- Matplotlib for visualization
- Model saved in HDF5 format (rice_lef_cnn.h5)